

## Supporting Information

### *When Does Generator Metadata Help? A Controlled Study of Structure-Aware Row Scaling for Generated Mixed-Integer Programs*

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This document contains (i) the 0.1%-MIPGap Gurobi distributional, diagnostic, and performance-profile views (Scenario 2 of the main text; Figures S1–S3), which mirror the corresponding 1%-MIPGap figures of the main text; (ii) the 0.1%-MIPGap CPLEX distributional and profile views (Scenario 4; Figure S4); (iii) the hyperparameter sensitivity of the structure-aware rule (Table S1); (iv) the per-variant original-scale reliability breakdown with PAR10 under the three readings of the main text (Table S2); (v) the out-of-sample sensitivity of the collective coupling reliability floor (Table S3); (vi) the evaluation of the optional local reliability floor on the failing Simple subset (Table S4); (vii) the common-pool exclusions and full-pool sensitivity (Table S5); (viii) paired-test attrition, the corrected Wilcoxon procedure, and clustered-bootstrap accounting (Table S6); (ix) the available solver-internal-scaling sweep and the verified CPLEX `Read::Scale= -1` audit (Table S7); and (x) the archived runtime comparison used for the coupling-stage activation audit (Table S8); (xi) the direct SA–Flat strict/mixed reliability audit (Table S9); and (xii) the multiplicity, operational-family, bootstrap, preprocessing-plus-solver, deterministic-effort, and own-incumbent sensitivities for that direct comparison (Table S10); (xiii) its per-contrast attrition and cached-metadata endpoint on both builds (Table S11); and (xiv) the measurement environment, build comparison and cross-environment transfer-factor sensitivity (Table S12); and (xv) the full traceability of the kernel-matched metadata grid—the block-heterogeneous family, its per-block scale distribution, the seeds, the per-variant  $\kappa_2^+$  distribution on the headline cell, and the reproduction command (Section S3, Table S13).

## Section S1. Proofs of the spectral propositions

These are the full proofs of the two propositions stated in Section 3 of the main text, moved here to keep the motivation and the method adjacent.

### **Proof of Proposition 1 (Ideal scalar scaling of block-diagonal matrices).**

*Proof.* For fixed positive scalars  $\alpha_i$ , the scaled matrix is block diagonal:

$$\text{diag}(\alpha_1 I, \dots, \alpha_B I) D = \text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B).$$

The singular values of a block-diagonal matrix are the union of the singular values of its blocks. Therefore, for every  $i$ ,

$$\kappa_2(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) \geq \kappa_2(\alpha_i A_i) = \kappa_2(A_i).$$

Taking the maximum over  $i$  gives the lower bound

$$\kappa_2(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) \geq \max_i \kappa_2(A_i)$$

for every admissible block scaling.

Now choose  $\alpha_i = (M_i m_i)^{-1/2}$ . Then the largest and smallest singular values of  $\alpha_i A_i$  are

$$\begin{aligned} \sigma_{\max}(\alpha_i A_i) &= \sqrt{\frac{M_i}{m_i}} = \sqrt{\kappa_2(A_i)}, \\ \sigma_{\min}(\alpha_i A_i) &= \sqrt{\frac{m_i}{M_i}} = \frac{1}{\sqrt{\kappa_2(A_i)}}. \end{aligned}$$

Because the singular values of the complete block-diagonal matrix are obtained by pooling the singular values of its blocks,

$$\sigma_{\max}(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) = \sqrt{\max_i \kappa_2(A_i)},$$

and

$$\sigma_{\min}(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) = \frac{1}{\sqrt{\max_i \kappa_2(A_i)}}.$$

The resulting condition number is  $\max_i \kappa_2(A_i)$ , matching the lower bound.  $\square$

**Proof of Proposition 2 (Conditioning of a triangular coupling factor).**

*Proof.* Let  $R = U\Sigma V^\top$  be a singular value decomposition (SVD) of  $R$ , with  $U$  and  $V$  orthogonal. Orthogonal multiplication does not change singular values, and hence  $T_R$  has the same singular values as

$$\begin{bmatrix} V^\top & 0 \\ 0 & U^\top \end{bmatrix} \begin{bmatrix} I_n & 0 \\ R & I_m \end{bmatrix} \begin{bmatrix} V & 0 \\ 0 & U \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ \Sigma & I_m \end{bmatrix}.$$

For each singular value  $\sigma$  of  $R$ , the nontrivial part reduces to the two-dimensional block

$$B_\sigma = \begin{bmatrix} 1 & 0 \\ \sigma & 1 \end{bmatrix}.$$

The squared singular values of  $B_\sigma$  are the eigenvalues of

$$B_\sigma^\top B_\sigma = \begin{bmatrix} 1 + \sigma^2 & \sigma \\ \sigma & 1 \end{bmatrix},$$

namely

$$\lambda_\pm(\sigma) = \frac{\sigma^2 + 2 \pm \sigma\sqrt{\sigma^2 + 4}}{2}.$$

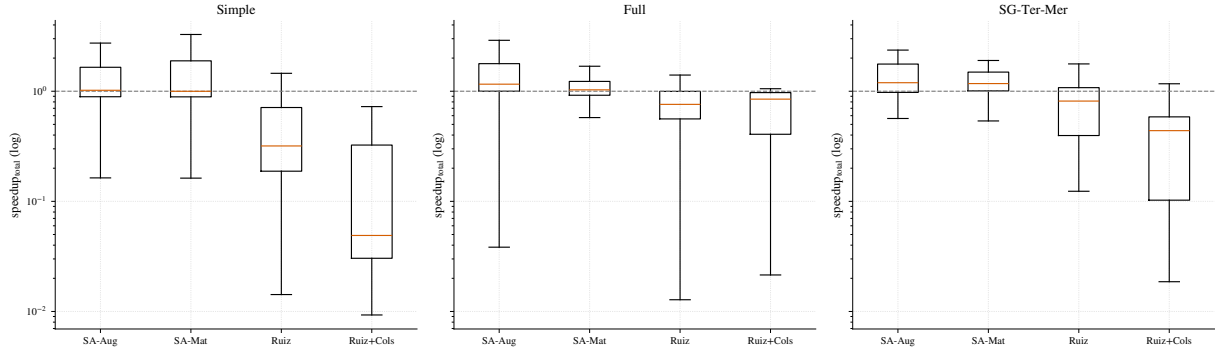
Their product is one, and  $\lambda_+(\sigma)$  is increasing for  $\sigma \geq 0$ . Additional directions associated with zero singular values of  $R$  contribute singular values equal to one. Therefore the largest squared singular value of  $T_R$  is  $\lambda_+(\rho)$  and the smallest squared singular value is  $1/\lambda_+(\rho)$ . Consequently,

$$\begin{aligned} \kappa_2(T_R) &= \lambda_+(\rho) \\ &= \frac{\rho^2 + 2 + \rho\sqrt{\rho^2 + 4}}{2} \\ &= \left( \frac{\sqrt{\rho^2 + 4} + \rho}{2} \right)^2. \end{aligned}$$

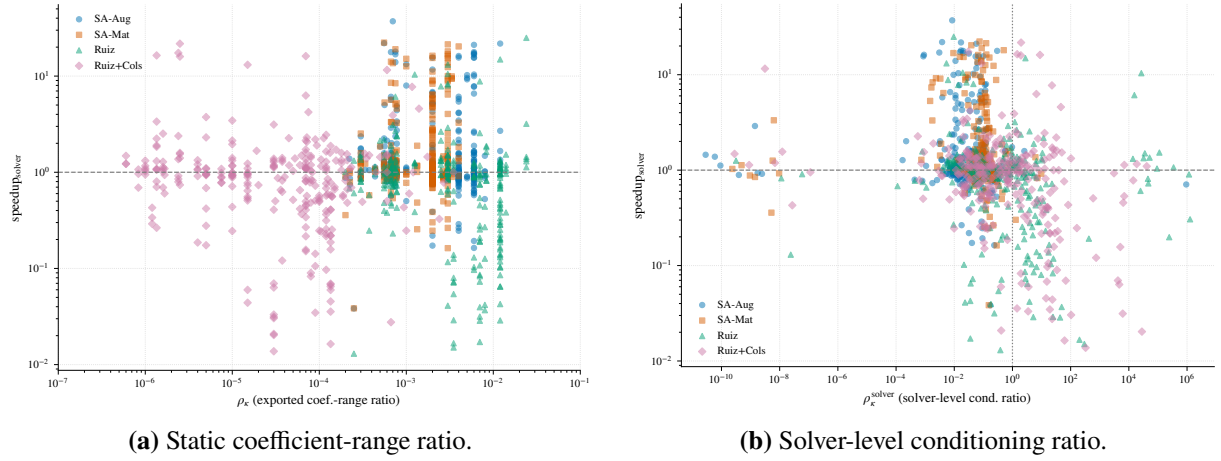
$\square$

## Section S2. Supplementary computational results

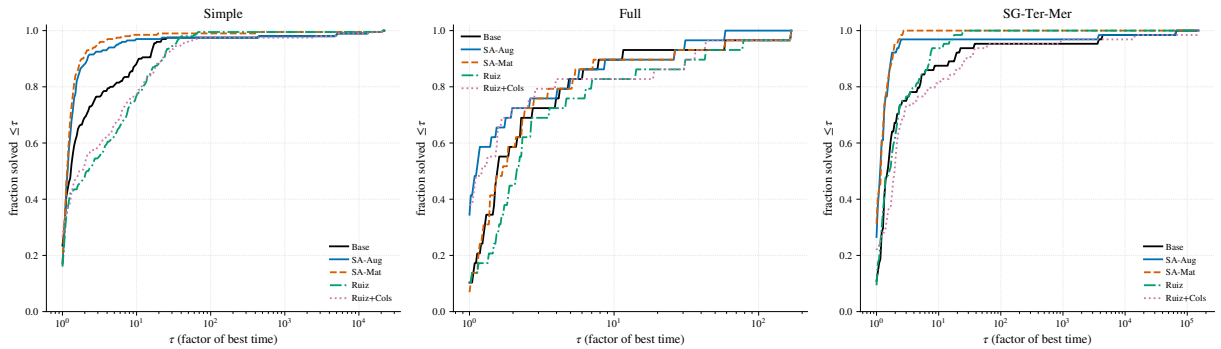
Here, PAR10 (including its ITT/strict/mixed superscripts) denotes the common-pool metric defined in the main text. The distinct symbol PAR10<sub>avail</sub> is reserved for archived descriptive revision tables: its denominator is the variant's available solver-output records, a timeout receives 36,000 s, and a pure pipeline error with no solver output is excluded. It must not be compared numerically with the fixed-pool metric.



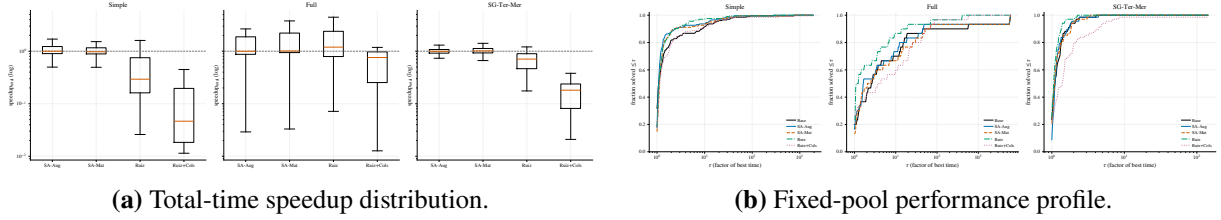
**Figure S1:** At 0.1% MIPGap, the Gurobi speedup distributions preserve the same qualitative pattern observed at the looser tolerance. Values above one indicate faster runs than the unscaled formulation.



**Figure S2:** At 0.1% MIPGap, the pooled scatter shows a stronger association with speedup for the own-incumbent path-specific fixed-LP diagnostic than for the exported-range proxy. This describes solver paths; it does not make the former a purer measure of representation or an instance-level predictor for the structure-aware variants.



**Figure S3:** At 0.1% MIPGap, fixed-pool Dolan–Moré profiles use the common denominators and individual 36,000-s failure cost defined in the main text.



**Figure S4:** At 0.1% MIPGap, the CPLEX visual evidence remains mixed under the stricter tolerance. The profile uses a common pool and assigns 36,000 s to an individual timeout, abort, missing run, or invalid time, as defined in the main text.

**Table S1:** Hyperparameter sensitivity of SA-Aug (Simple class, Gurobi, 1% MIPGap). Solver-time sgm ( $s = 1$  s) and descriptive  $\text{PAR10}_{\text{avail}}$ , in seconds; default settings in bold.

| Parameter                                         | Value                               | sgm (s) | $\text{PAR10}_{\text{avail}}$ (s) |
|---------------------------------------------------|-------------------------------------|---------|-----------------------------------|
| Band $u$                                          | 1.5                                 | 2.09    | 2.46                              |
|                                                   | <b>2</b>                            | 2.07    | 2.45                              |
|                                                   | 4                                   | 2.12    | 2.49                              |
| Window $[\varepsilon_{\min}, \varepsilon_{\max}]$ | $[10^{-6}, 10^6]$                   | 2.04    | 2.40                              |
|                                                   | <b><math>[10^{-9}, 10^9]</math></b> | 2.02    | 2.39                              |
|                                                   | $[10^{-12}, 10^{12}]$               | 2.01    | 2.36                              |

Base sgm is 3.73 s for reference. Each row is a separate matched run; the spread across all settings ( $\text{sgm} \in [2.01, 2.12]$ ) is far below the base-to-scaled gap, and the median paired speedup is  $\approx 1.0$  throughout. The number of Ruiz iterations  $K$  is a parameter of the Ruiz baselines, not of the structure-aware rule, so it is fixed at  $K = 4$ ; varying it leaves SA-Aug unchanged, as expected.

**Table S2:** Per-variant audit and PAR10 under the three operational readings. All campaigns use the manifest-declared  $T_{\max} = 3600$  s. A timeout or pipeline abort receives  $10T_{\max} = 36000$  s in every reading; a returned output that fails the relevant verifier receives that penalty only in the strict or mixed reading. The count columns use the raw manifest pool:  $n_{\text{att}}$  is every attempted instance and  $n_{\text{out}}$  reproduces the pooled diagnostic audit’s status-OK denominator, including time-limited runs and three records that lack verification fields. TO is a subset of  $n_{\text{out}}$ , and Abort is  $n_{\text{att}} - n_{\text{out}}$ . Pass counts are among those status-OK records; missing verification is non-passing. Strict pass means  $v_{\text{row}} \leq 10^{-4}$  and  $v_{\text{int}} \leq 10^{-5}$ ; mixed pass rescues only row failures with  $v_r/(1 + |b_r|) \leq 10^{-6}$ . In contrast, the four PAR10 columns use the reproducible scenario-specific common pool after the uniformly-unavailable exclusion in Table S5: Base ITT and, for the listed variant, ITT/strict/mixed. Values are seconds, rounded to the nearest second.

| Solver | Gap  | Class      | Variant   | $n_{\text{att}}$ | $n_{\text{out}}$ | TO | Abort | $n_{\text{pass}}^{\text{str}}$ | $n_{\text{pass}}^{\text{mix}}$ | Base ITT | ITT  | Strict | Mixed |
|--------|------|------------|-----------|------------------|------------------|----|-------|--------------------------------|--------------------------------|----------|------|--------|-------|
| Gurobi | 1%   | Simple     | SA-Aug    | 204              | 197              | 0  | 7     | 185                            | 185                            | 373      | 364  | 2535   | 2535  |
| Gurobi | 1%   | Simple     | SA-Mat    | 204              | 198              | 0  | 6     | 179                            | 180                            | 373      | 183  | 3620   | 3439  |
| Gurobi | 1%   | Simple     | Ruiz      | 204              | 198              | 0  | 6     | 143                            | 146                            | 373      | 207  | 10153  | 9611  |
| Gurobi | 1%   | Simple     | Ruiz+Cols | 204              | 199              | 0  | 5     | 180                            | 189                            | 373      | 28   | 3462   | 1835  |
| Gurobi | 1%   | SG-Ter-Mer | SA-Aug    | 68               | 67               | 1  | 1     | 66                             | 67                             | 1094     | 1083 | 1613   | 1083  |
| Gurobi | 1%   | SG-Ter-Mer | SA-Mat    | 68               | 67               | 0  | 1     | 65                             | 66                             | 1094     | 1079 | 1608   | 1079  |
| Gurobi | 1%   | SG-Ter-Mer | Ruiz      | 68               | 68               | 1  | 0     | 67                             | 68                             | 1094     | 567  | 1095   | 567   |
| Gurobi | 1%   | SG-Ter-Mer | Ruiz+Cols | 68               | 68               | 1  | 0     | 67                             | 68                             | 1094     | 585  | 1113   | 585   |
| Gurobi | 1%   | Full       | SA-Aug    | 34               | 34               | 3  | 0     | 27                             | 27                             | 3385     | 3308 | 9640   | 9640  |
| Gurobi | 1%   | Full       | SA-Mat    | 34               | 34               | 2  | 0     | 31                             | 31                             | 3385     | 3473 | 5586   | 5586  |
| Gurobi | 1%   | Full       | Ruiz      | 34               | 33               | 4  | 1     | 31                             | 31                             | 3385     | 5491 | 7577   | 7577  |
| Gurobi | 1%   | Full       | Ruiz+Cols | 34               | 33               | 4  | 1     | 31                             | 32                             | 3385     | 5617 | 6666   | 5617  |
| Gurobi | 0.1% | Simple     | SA-Aug    | 204              | 195              | 0  | 9     | 174                            | 178                            | 928      | 928  | 4705   | 3987  |
| Gurobi | 0.1% | Simple     | SA-Mat    | 204              | 198              | 0  | 6     | 168                            | 172                            | 928      | 381  | 5776   | 5057  |
| Gurobi | 0.1% | Simple     | Ruiz      | 204              | 199              | 0  | 5     | 155                            | 159                            | 928      | 221  | 8138   | 7419  |
| Gurobi | 0.1% | Simple     | Ruiz+Cols | 204              | 195              | 0  | 9     | 175                            | 185                            | 928      | 948  | 4544   | 2746  |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Aug    | 68               | 63               | 1  | 5     | 62                             | 63                             | 2246     | 1683 | 2237   | 1683  |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Mat    | 68               | 65               | 0  | 3     | 63                             | 64                             | 2246     | 576  | 1130   | 576   |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz      | 68               | 65               | 1  | 3     | 65                             | 65                             | 2246     | 587  | 587    | 587   |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz+Cols | 68               | 62               | 1  | 6     | 61                             | 62                             | 2246     | 2262 | 2814   | 2262  |
| Gurobi | 0.1% | Full       | SA-Aug    | 34               | 30               | 3  | 4     | 27                             | 27                             | 5992     | 5897 | 8140   | 8140  |
| Gurobi | 0.1% | Full       | SA-Mat    | 34               | 31               | 4  | 3     | 28                             | 30                             | 5992     | 6026 | 9293   | 7148  |
| Gurobi | 0.1% | Full       | Ruiz      | 34               | 30               | 4  | 4     | 27                             | 27                             | 5992     | 7066 | 9277   | 9277  |
| Gurobi | 0.1% | Full       | Ruiz+Cols | 34               | 31               | 5  | 3     | 28                             | 29                             | 5992     | 7073 | 9312   | 8191  |
| CPLEX  | 1%   | Simple     | SA-Aug    | 204              | 204              | 0  | 0     | 37                             | 37                             | 6        | 3    | 29471  | 29471 |
| CPLEX  | 1%   | Simple     | SA-Mat    | 204              | 204              | 0  | 0     | 43                             | 43                             | 6        | 3    | 28412  | 28412 |
| CPLEX  | 1%   | Simple     | Ruiz      | 204              | 204              | 0  | 0     | 44                             | 44                             | 6        | 4    | 28237  | 28237 |
| CPLEX  | 1%   | Simple     | Ruiz+Cols | 204              | 204              | 0  | 0     | 137                            | 138                            | 6        | 3    | 11825  | 11649 |
| CPLEX  | 1%   | SG-Ter-Mer | SA-Aug    | 68               | 68               | 0  | 0     | 68                             | 68                             | 3        | 10   | 10     | 10    |
| CPLEX  | 1%   | SG-Ter-Mer | SA-Mat    | 68               | 68               | 0  | 0     | 68                             | 68                             | 3        | 6    | 6      | 6     |
| CPLEX  | 1%   | SG-Ter-Mer | Ruiz      | 68               | 68               | 0  | 0     | 68                             | 68                             | 3        | 10   | 10     | 10    |
| CPLEX  | 1%   | SG-Ter-Mer | Ruiz+Cols | 68               | 68               | 0  | 0     | 68                             | 68                             | 3        | 15   | 15     | 15    |

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Table S2 (continued)

| Solver | Gap  | Class      | Variant   | $n_{\text{att}}$ | $n_{\text{out}}$ | TO | Abort | $n_{\text{pass}}^{\text{str}}$ | $n_{\text{pass}}^{\text{mix}}$ | Base ITT | ITT   | Strict | Mixed |
|--------|------|------------|-----------|------------------|------------------|----|-------|--------------------------------|--------------------------------|----------|-------|--------|-------|
| CPLEX  | 1%   | Full       | SA-Aug    | 34               | 34               | 5  | 0     | 25                             | 25                             | 6515     | 5546  | 15014  | 15014 |
| CPLEX  | 1%   | Full       | SA-Mat    | 34               | 34               | 6  | 0     | 25                             | 26                             | 6515     | 6498  | 14932  | 13875 |
| CPLEX  | 1%   | Full       | Ruiz      | 34               | 34               | 3  | 0     | 28                             | 28                             | 6515     | 3506  | 8744   | 8744  |
| CPLEX  | 1%   | Full       | Ruiz+Cols | 34               | 34               | 4  | 0     | 32                             | 33                             | 6515     | 4681  | 6793   | 5737  |
| CPLEX  | 0.1% | Simple     | SA-Aug    | 204              | 204              | 1  | 0     | 40                             | 40                             | 214      | 188   | 29119  | 29119 |
| CPLEX  | 0.1% | Simple     | SA-Mat    | 204              | 204              | 1  | 0     | 38                             | 38                             | 214      | 203   | 29311  | 29311 |
| CPLEX  | 0.1% | Simple     | Ruiz      | 204              | 204              | 1  | 0     | 44                             | 44                             | 214      | 192   | 28413  | 28413 |
| CPLEX  | 0.1% | Simple     | Ruiz+Cols | 204              | 204              | 3  | 0     | 144                            | 146                            | 214      | 547   | 10955  | 10779 |
| CPLEX  | 0.1% | SG-Ter-Mer | SA-Aug    | 68               | 68               | 0  | 0     | 68                             | 68                             | 19       | 21    | 21     | 21    |
| CPLEX  | 0.1% | SG-Ter-Mer | SA-Mat    | 68               | 68               | 0  | 0     | 68                             | 68                             | 19       | 16    | 16     | 16    |
| CPLEX  | 0.1% | SG-Ter-Mer | Ruiz      | 68               | 68               | 0  | 0     | 68                             | 68                             | 19       | 8     | 8      | 8     |
| CPLEX  | 0.1% | SG-Ter-Mer | Ruiz+Cols | 68               | 68               | 1  | 0     | 68                             | 68                             | 19       | 539   | 539    | 539   |
| CPLEX  | 0.1% | Full       | SA-Aug    | 34               | 34               | 8  | 0     | 24                             | 24                             | 9951     | 8930  | 18283  | 18283 |
| CPLEX  | 0.1% | Full       | SA-Mat    | 34               | 33               | 10 | 1     | 25                             | 26                             | 9951     | 11916 | 20280  | 19226 |
| CPLEX  | 0.1% | Full       | Ruiz      | 34               | 34               | 5  | 0     | 32                             | 32                             | 9951     | 5857  | 7917   | 7917  |
| CPLEX  | 0.1% | Full       | Ruiz+Cols | 34               | 33               | 10 | 1     | 31                             | 32                             | 9951     | 11950 | 14063  | 13007 |

*Objective-difference audit.* MIPGap bounds each incumbent against its own dual bound and is not a cross-run agreement test. Of 4,806 variant runs with returned solver output, 72 differ from the corresponding Base incumbent by more than twice the nominal gap; these cases are concentrated in a small set of instance identifiers. The largest pooled relative deviations occur in Full/Ruiz cells ( $|\Delta z_{\text{rel}}| \approx 1.8 \times 10^2$  under Gurobi and  $5.3 \times 10^1$  under CPLEX). Base is not presumed correct: one Simple Base incumbent is 73% worse than the best verified incumbent, and one CPLEX Full Base incumbent is roughly six times worse. Accordingly, correctness is assessed against the best original-scale-verified incumbent on each instance, with Base subjected to the same feasibility and integrality checks as every scaled variant.

**Table S3:** Out-of-sample sensitivity of the reliability floor for the collective coupling factor (SA-Aug), on twenty seeds (21–40) never used while designing the floor. Cells: coupling, coupling-uniform, and mixed patterns at severity  $S = 6$  (60 instances, Gurobi, MIPGap  $10^{-3}$ , 600 s limit). Median post-scaling global coefficient-and-right-hand-side magnitude-range proxy per pattern; number of runs failing original-scale verification; worst original-scale row violation; instances whose objective disagrees with the base beyond the MIPGap. The failure mode replicates out-of-sample with the floor disabled, and every floor value across three orders of magnitude eliminates it.

| Floor               | coupling          | coupling-unif.    | mixed                | $n_{\text{fail}}$ | $v_{\text{max}}$     | over gap |
|---------------------|-------------------|-------------------|----------------------|-------------------|----------------------|----------|
| (pre-scaling)       | $4.4 \times 10^8$ | $3.3 \times 10^6$ | $1.5 \times 10^{14}$ |                   |                      |          |
| disabled            | $3.8 \times 10^7$ | $5.4 \times 10^3$ | $2.9 \times 10^7$    | 4                 | $4.2 \times 10^{-1}$ | 4        |
| $10^{-2}$           | $9.1 \times 10^5$ | $5.4 \times 10^3$ | $2.9 \times 10^6$    | 0                 | $1.2 \times 10^{-7}$ | 0        |
| $10^{-3}$ (default) | $3.5 \times 10^5$ | $5.4 \times 10^3$ | $2.3 \times 10^6$    | 0                 | $1.2 \times 10^{-7}$ | 0        |
| $10^{-4}$           | $1.0 \times 10^6$ | $5.4 \times 10^3$ | $2.3 \times 10^6$    | 0                 | $1.2 \times 10^{-7}$ | 0        |
| $10^{-5}$           | $2.1 \times 10^6$ | $5.4 \times 10^3$ | $3.0 \times 10^6$    | 0                 | $1.2 \times 10^{-7}$ | 0        |

*Coupling-trigger sensitivity.* On the original synthetic grid, sweeping  $\hat{\rho}_{\text{trigger}} \in \{2, 5, 10, 20, 50\}$  leaves every displayed coupling and coupling-uniform cell median unchanged. Approximately one instance in twenty lies in the  $[2, 50]$  band, so some individual activation decisions change, but the proxy is effectively bimodal (above  $10^2$  on oversized instances and below one on undersized instances). The result therefore supports the stated conclusion about the one-sided design, not a privileged choice of the cutoff 10.

**Table S4:** Optional local reliability floor (SA-Aug-LF), evaluated on the twelve Simple instances (Gurobi, 1% MIPGap) whose SA-Aug runs fail original-scale verification. The floor applies the local factor only if  $\alpha_r^{A,b} \geq \varepsilon_{\text{floor}}^{\text{loc}}$ , so  $d_r \geq \varepsilon_{\text{floor}}^{\text{loc}}$  bounds the conditional original-scale residual by  $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$ ; rows below the floor stay at base scale.  $n_{\text{fail}}$  is the number of the twelve runs failing the strict absolute criterion;  $\sigma_{\text{gm}}$  is the shifted geometric mean solver time ( $s = 1$  s);  $\hat{\kappa}_{\text{range}}^{\text{post}}$  is the median exported coefficient-and-right-hand-side magnitude-range proxy;  $v_{\text{max}}$  is the worst original-scale row violation. SA-Mat is shown for reference.

| Variant                                                        | $n_{\text{fail}}$ | $\sigma_{\text{gm}}$ (s) | $\hat{\kappa}_{\text{range}}^{\text{post}}$ | $v_{\text{max}}$     |
|----------------------------------------------------------------|-------------------|--------------------------|---------------------------------------------|----------------------|
| Base                                                           | 0                 | 4.90                     | —                                           | 0                    |
| SA-Aug (no floor, default)                                     | 12                | 2.47                     | $6.6 \times 10^9$                           | $3.7 \times 10^{-3}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-1}$ | 0                 | 2.64                     | $5.5 \times 10^{11}$                        | $5.3 \times 10^{-5}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-2}$ | 0                 | 2.89                     | $5.5 \times 10^{11}$                        | $7.4 \times 10^{-5}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-3}$ | 6                 | 5.52                     | $2.0 \times 10^{14}$                        | $4.0 \times 10^{-2}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-4}$ | 6                 | 2.19                     | $2.0 \times 10^{14}$                        | $5.3 \times 10^{-4}$ |
| SA-Mat                                                         | 5                 | 2.57                     | $4.5 \times 10^{12}$                        | $1.9 \times 10^{-3}$ |

*Notes.* The floor is monotone:  $n_{\text{fail}}$  falls from 12 (no floor) to 6 (shallow floors  $10^{-3}/10^{-4}$ , which protect only the most extreme rows) to 0 ( $10^{-2}/10^{-1}$ ), consistent with the residual bound  $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$ . At the effective floor  $10^{-2}$  the variant eliminates every failure while retaining a solver-time advantage over the base ( $\sigma_{\text{gm}}$  2.89 vs 4.90 s), even though it *raises* the exported magnitude-range proxy (the protected rows are exactly the over-compressed large-right-hand-side rows)—a further illustration that this static proxy is a weak predictor of runtime. The threshold is imposed on the representation-dependent factor  $\alpha_r^{A,b}$  and is therefore not invariant to a positive re-emission of the same unscaled row. Here it is defined relative to the generator’s canonical export in physical units. A representation-invariant alternative could threshold a norm of the resulting scaled row, but it would not inherit the same  $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$  residual bound. The  $\sigma_{\text{gm}}$  values on twelve instances are noisy and are not a full-class estimate. This subset was selected by a post-treatment outcome (the SA-Aug failures), so the table is a demonstration on the failing set; the full-class reliability and runtime effect, and the behavior on Full, remain untested.

**Table S5:** Uniformly unavailable operational records and sensitivity to retaining them. Panel A applies the documented reproducible data rule: exclude an instance only when all five attempted variants have nonzero pipeline return codes. Panel B restores those cells to the manifest-declared full pool and assigns each a 36000-s penalty. Triplets are ITT/strict/mixed PAR10 in seconds; only scenarios with exclusions are shown because common- and full-pool results otherwise coincide. The within-reading ordering of Base, SA-Aug, and SA-Mat is unchanged (the rounded common-pool tie in Gurobi Simple at 0.1% is broken by less than one second before rounding in the full pool).

| Panel A: identifiers |      |            |         |                                   |
|----------------------|------|------------|---------|-----------------------------------|
| Solver               | Gap  | Class      | $N/n$   | Uniformly unavailable identifiers |
| Gurobi               | 1%   | Simple     | 204/199 | caso46f, caso47a–caso47d          |
| Gurobi               | 0.1% | Simple     | 204/200 | caso02c–caso02e, caso22a          |
| Gurobi               | 0.1% | SG-Ter-Mer | 68/65   | instance020–instance022           |
| Gurobi               | 0.1% | Full       | 34/32   | instance046, instance047          |

**Panel B: common pool versus penalized full pool**

| Gap/class                                   | Variant   | Pool   | ITT  | Strict | Mixed |
|---------------------------------------------|-----------|--------|------|--------|-------|
| 1%, Simple<br>( $n = 199$ , $N = 204$ )     | Base      | common | 373  | 554    | 554   |
|                                             |           | full   | 1246 | 1422   | 1422  |
|                                             | SA-Aug    | common | 364  | 2535   | 2535  |
|                                             |           | full   | 1238 | 3355   | 3355  |
|                                             | SA-Mat    | common | 183  | 3620   | 3439  |
|                                             |           | full   | 1061 | 4414   | 4237  |
|                                             | Ruiz      | common | 207  | 10153  | 9611  |
|                                             |           | full   | 1085 | 10786  | 10258 |
|                                             | Ruiz+Cols | common | 28   | 3462   | 1835  |
|                                             |           | full   | 909  | 4260   | 2673  |
| 0.1%, Simple<br>( $n = 200$ , $N = 204$ )   | Base      | common | 928  | 1108   | 1108  |
|                                             |           | full   | 1615 | 1792   | 1792  |
|                                             | SA-Aug    | common | 928  | 4705   | 3987  |
|                                             |           | full   | 1616 | 5318   | 4615  |
|                                             | SA-Mat    | common | 381  | 5776   | 5057  |
|                                             |           | full   | 1079 | 6369   | 5664  |
|                                             | Ruiz      | common | 221  | 8138   | 7419  |
|                                             |           | full   | 923  | 8684   | 7980  |
|                                             | Ruiz+Cols | common | 948  | 4544   | 2746  |
|                                             |           | full   | 1635 | 5161   | 3398  |
| 0.1%, SG-Ter-Mer<br>( $n = 65$ , $N = 68$ ) | Base      | common | 2246 | 2246   | 2246  |
|                                             |           | full   | 3735 | 3735   | 3735  |
|                                             | SA-Aug    | common | 1683 | 2237   | 1683  |
|                                             |           | full   | 3197 | 3727   | 3197  |
|                                             | SA-Mat    | common | 576  | 1130   | 576   |
|                                             |           | full   | 2139 | 2668   | 2139  |
|                                             | Ruiz      | common | 587  | 587    | 587   |
|                                             |           | full   | 2149 | 2149   | 2149  |
|                                             | Ruiz+Cols | common | 2262 | 2814   | 2262  |
|                                             |           | full   | 3750 | 4279   | 3750  |
| 0.1%, Full<br>( $n = 32$ , $N = 34$ )       | Base      | common | 5992 | 5992   | 5992  |
|                                             |           | full   | 7757 | 7757   | 7757  |
|                                             | SA-Aug    | common | 5897 | 8140   | 8140  |
|                                             |           | full   | 7668 | 9779   | 9779  |
|                                             | SA-Mat    | common | 6026 | 9293   | 7148  |
|                                             |           | full   | 7789 | 10864  | 8845  |
|                                             | Ruiz      | common | 7066 | 9277   | 9277  |
|                                             |           | full   | 8768 | 10849  | 10849 |
|                                             | Ruiz+Cols | common | 7073 | 9312   | 8191  |
|                                             |           | full   | 8775 | 10882  | 9827  |

*Note.*  $N$  is the manifest pool and  $n$  the common pool. In the archived logs for every excluded identifier, Base, SA-Aug, Ruiz, and Ruiz+Cols stop at a Gurobi WLS `Overage for too long` licensing error and report no configured solver. The SA-Mat logs instead record optimal completions written to a separate `*-matricial` result tree, while their aggregate campaign records retain nonzero return codes. Thus the evidence supports a combination of license-capacity failure and incomplete artifact consolidation; the exclusion itself is defined only by the observable all-variant return-code rule and does not assert that every underlying solve failed.

**Table S6:** Pair attrition and corrected Wilcoxon audit.  $n$  is the effective matched-pair count. UM denotes the all-variant uniformly-unavailable exclusion;  $D_{TO}$ ,  $D_A$ , and  $D_{mix}$  are exclusions because both Base and the listed variant are, respectively, timeouts, pipeline aborts, or different kinds of noncompletion.  $C_1$  is a one-sided noncompletion retained at exactly  $T_{max} = 3600$  s. There are no missing-time or exact-zero pairs in any row. A one-sided timeout is right-censored at  $T_{max}$ ; assigning the same cost to a one-sided algorithmic/pipeline abort is an operational-failure convention rather than a censoring claim. The test is two-sided Wilcoxon signed-rank on  $t_{variant} - t_{Base}$  (SciPy 1.15.3), with exact zeros discarded, tied absolute differences assigned midranks, and no continuity correction. The exact null distribution is used for at most 50 nonzero untied differences; otherwise the tie-corrected normal approximation is used. The displayed structure-aware Simple/SG rows use the approximation and Full rows use the exact distribution; no zero or tied-absolute-difference pair occurs.  $r_{rb} < 0$  favors the listed variant.  $q$  is Benjamini–Hochberg adjusted over the twelve tests in each solver-by-gap family (three classes by four variants).

| Solver | Gap  | Class      | Variant   | $n$ | UM | $D_{TO}$ | $D_A$ | $D_{mix}$ | $C_1$ | $W$  | $p$                    | $q$                    | $r_{rb}$ |
|--------|------|------------|-----------|-----|----|----------|-------|-----------|-------|------|------------------------|------------------------|----------|
| Gurobi | 1%   | Simple     | SA-Aug    | 197 | 5  | 0        | 2     | 0         | 0     | 7535 | $5.67 \times 10^{-3}$  | $9.72 \times 10^{-3}$  | −0.227   |
| Gurobi | 1%   | Simple     | SA-Mat    | 198 | 5  | 0        | 1     | 0         | 1     | 3011 | $2.42 \times 10^{-17}$ | $2.90 \times 10^{-16}$ | −0.694   |
| Gurobi | 1%   | Simple     | Ruiz      | 198 | 5  | 0        | 1     | 0         | 1     | 7248 | $1.27 \times 10^{-3}$  | $3.04 \times 10^{-3}$  | +0.264   |
| Gurobi | 1%   | Simple     | Ruiz+Cols | 199 | 5  | 0        | 0     | 0         | 2     | 7414 | $1.82 \times 10^{-3}$  | $3.65 \times 10^{-3}$  | +0.255   |
| Gurobi | 1%   | SG-Ter-Mer | SA-Aug    | 66  | 0  | 1        | 1     | 0         | 0     | 501  | $1.13 \times 10^{-4}$  | $4.51 \times 10^{-4}$  | −0.547   |
| Gurobi | 1%   | SG-Ter-Mer | SA-Mat    | 66  | 0  | 0        | 1     | 1         | 0     | 115  | $2.49 \times 10^{-10}$ | $1.50 \times 10^{-9}$  | −0.896   |
| Gurobi | 1%   | SG-Ter-Mer | Ruiz      | 67  | 0  | 1        | 0     | 0         | 1     | 1129 | 0.950                  | 0.950                  | −0.009   |
| Gurobi | 1%   | SG-Ter-Mer | Ruiz+Cols | 67  | 0  | 1        | 0     | 0         | 1     | 598  | $7.26 \times 10^{-4}$  | $2.18 \times 10^{-3}$  | +0.475   |
| Gurobi | 1%   | Full       | SA-Aug    | 32  | 0  | 2        | 0     | 0         | 2     | 175  | 0.0984                 | 0.148                  | −0.337   |
| Gurobi | 1%   | Full       | SA-Mat    | 32  | 0  | 2        | 0     | 0         | 2     | 204  | 0.270                  | 0.360                  | −0.227   |
| Gurobi | 1%   | Full       | Ruiz      | 31  | 0  | 2        | 0     | 1         | 2     | 197  | 0.327                  | 0.393                  | +0.206   |
| Gurobi | 1%   | Full       | Ruiz+Cols | 32  | 0  | 2        | 0     | 0         | 4     | 243  | 0.705                  | 0.769                  | +0.080   |
| Gurobi | 0.1% | Simple     | SA-Aug    | 195 | 4  | 0        | 5     | 0         | 0     | 6963 | $1.02 \times 10^{-3}$  | $3.06 \times 10^{-3}$  | −0.271   |
| Gurobi | 0.1% | Simple     | SA-Mat    | 198 | 4  | 0        | 2     | 0         | 3     | 7069 | $5.70 \times 10^{-4}$  | $2.28 \times 10^{-3}$  | −0.282   |
| Gurobi | 0.1% | Simple     | Ruiz      | 199 | 4  | 0        | 1     | 0         | 4     | 7429 | $1.94 \times 10^{-3}$  | $4.66 \times 10^{-3}$  | +0.253   |
| Gurobi | 0.1% | Simple     | Ruiz+Cols | 199 | 4  | 0        | 1     | 0         | 8     | 7517 | $2.78 \times 10^{-3}$  | $5.46 \times 10^{-3}$  | +0.245   |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Aug    | 62  | 3  | 1        | 2     | 0         | 1     | 456  | $2.63 \times 10^{-4}$  | $1.58 \times 10^{-3}$  | −0.533   |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Mat    | 64  | 3  | 0        | 0     | 1         | 3     | 341  | $2.95 \times 10^{-6}$  | $3.53 \times 10^{-5}$  | −0.672   |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz      | 64  | 3  | 1        | 0     | 0         | 3     | 889  | 0.313                  | 0.341                  | −0.145   |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz+Cols | 64  | 3  | 1        | 0     | 0         | 6     | 599  | $3.19 \times 10^{-3}$  | $5.46 \times 10^{-3}$  | +0.424   |
| Gurobi | 0.1% | Full       | SA-Aug    | 28  | 2  | 3        | 1     | 0         | 2     | 124  | 0.0735                 | 0.110                  | −0.389   |
| Gurobi | 0.1% | Full       | SA-Mat    | 27  | 2  | 4        | 1     | 0         | 0     | 173  | 0.714                  | 0.714                  | −0.085   |
| Gurobi | 0.1% | Full       | Ruiz      | 28  | 2  | 3        | 0     | 1         | 3     | 144  | 0.186                  | 0.248                  | +0.291   |
| Gurobi | 0.1% | Full       | Ruiz+Cols | 28  | 2  | 4        | 0     | 0         | 3     | 149  | 0.227                  | 0.272                  | −0.266   |
| CPLEX  | 1%   | Simple     | SA-Aug    | 204 | 0  | 0        | 0     | 0         | 0     | 9377 | 0.202                  | 0.346                  | −0.103   |
| CPLEX  | 1%   | Simple     | SA-Mat    | 204 | 0  | 0        | 0     | 0         | 0     | 7460 | $3.89 \times 10^{-4}$  | $2.33 \times 10^{-3}$  | +0.286   |
| CPLEX  | 1%   | Simple     | Ruiz      | 204 | 0  | 0        | 0     | 0         | 0     | 9496 | 0.256                  | 0.384                  | +0.092   |
| CPLEX  | 1%   | Simple     | Ruiz+Cols | 204 | 0  | 0        | 0     | 0         | 0     | 8958 | 0.0762                 | 0.183                  | +0.143   |
| CPLEX  | 1%   | SG-Ter-Mer | SA-Aug    | 68  | 0  | 0        | 0     | 0         | 0     | 1068 | 0.521                  | 0.521                  | +0.090   |
| CPLEX  | 1%   | SG-Ter-Mer | SA-Mat    | 68  | 0  | 0        | 0     | 0         | 0     | 740  | $8.15 \times 10^{-3}$  | 0.0326                 | +0.369   |

Continued on next page

Table S6 (continued)

| Solver | Gap  | Class      | Variant   | $n$ | UM | $D_{TO}$ | $D_A$ | $D_{mix}$ | $C_1$ | $W$   | $p$                   | $q$                   | $r_{tb}$ |
|--------|------|------------|-----------|-----|----|----------|-------|-----------|-------|-------|-----------------------|-----------------------|----------|
| CPLEX  | 1%   | SG-Ter-Mer | Ruiz      | 68  | 0  | 0        | 0     | 0         | 0     | 1041  | 0.420                 | 0.458                 | +0.113   |
| CPLEX  | 1%   | SG-Ter-Mer | Ruiz+Cols | 68  | 0  | 0        | 0     | 0         | 0     | 438   | $7.09 \times 10^{-6}$ | $8.50 \times 10^{-5}$ | +0.627   |
| CPLEX  | 1%   | Full       | SA-Aug    | 29  | 0  | 4        | 0     | 1         | 1     | 172   | 0.336                 | 0.428                 | +0.209   |
| CPLEX  | 1%   | Full       | SA-Mat    | 28  | 0  | 5        | 0     | 1         | 0     | 131   | 0.104                 | 0.208                 | -0.355   |
| CPLEX  | 1%   | Full       | Ruiz      | 31  | 0  | 2        | 0     | 1         | 3     | 200   | 0.357                 | 0.428                 | -0.194   |
| CPLEX  | 1%   | Full       | Ruiz+Cols | 30  | 0  | 3        | 0     | 1         | 2     | 130   | 0.0345                | 0.104                 | +0.441   |
| CPLEX  | 0.1% | Simple     | SA-Aug    | 204 | 0  | 0        | 0     | 0         | 2     | 8765  | 0.0453                | 0.121                 | -0.162   |
| CPLEX  | 0.1% | Simple     | SA-Mat    | 203 | 0  | 1        | 0     | 0         | 0     | 10056 | 0.723                 | 0.723                 | -0.029   |
| CPLEX  | 0.1% | Simple     | Ruiz      | 204 | 0  | 0        | 0     | 0         | 2     | 9364  | 0.196                 | 0.356                 | -0.104   |
| CPLEX  | 0.1% | Simple     | Ruiz+Cols | 203 | 0  | 1        | 0     | 0         | 2     | 8092  | $6.98 \times 10^{-3}$ | 0.0419                | +0.218   |
| CPLEX  | 0.1% | SG-Ter-Mer | SA-Aug    | 68  | 0  | 0        | 0     | 0         | 0     | 1085  | 0.591                 | 0.709                 | +0.075   |
| CPLEX  | 0.1% | SG-Ter-Mer | SA-Mat    | 68  | 0  | 0        | 0     | 0         | 0     | 985   | 0.251                 | 0.372                 | -0.160   |
| CPLEX  | 0.1% | SG-Ter-Mer | Ruiz      | 68  | 0  | 0        | 0     | 0         | 0     | 853   | 0.0505                | 0.121                 | -0.273   |
| CPLEX  | 0.1% | SG-Ter-Mer | Ruiz+Cols | 68  | 0  | 0        | 0     | 0         | 1     | 473   | $1.89 \times 10^{-5}$ | $2.27 \times 10^{-4}$ | +0.597   |
| CPLEX  | 0.1% | Full       | SA-Aug    | 27  | 0  | 6        | 0     | 1         | 3     | 173   | 0.714                 | 0.723                 | -0.085   |
| CPLEX  | 0.1% | Full       | SA-Mat    | 26  | 0  | 6        | 0     | 2         | 4     | 125   | 0.208                 | 0.356                 | -0.288   |
| CPLEX  | 0.1% | Full       | Ruiz      | 30  | 0  | 3        | 0     | 1         | 6     | 134   | 0.0427                | 0.121                 | -0.424   |
| CPLEX  | 0.1% | Full       | Ruiz+Cols | 27  | 0  | 6        | 0     | 1         | 6     | 143   | 0.279                 | 0.372                 | +0.243   |

II

*One-sided-abort sensitivity.* Replacing the primary  $T_{\max}$  convention by  $10T_{\max}$  for a one-sided algorithmic/pipeline abort leaves every effect direction and inferential decision unchanged. The largest change among the displayed Gurobi structure-aware rows is Full/SA-Aug at 0.1%:  $W = 124 \rightarrow 125$ ,  $p = 0.07354 \rightarrow 0.07742$ ,  $q_{BH} = 0.11031 \rightarrow 0.11612$ , and  $r_{tb} = -0.389 \rightarrow -0.384$ . All other Gurobi structure-aware rows are unchanged; CPLEX structure-aware  $W$ ,  $p$ , and  $r_{tb}$  are unchanged, with only minor BH shifts caused by other members of the twelve-test family. The sensitivity is reproduced by `scripts/paired_tests.py --abort-penalty par10`.

*Cluster-bootstrap accounting for the conditioning comparison.* At 1%, 298 of 306 instances have at least one complete conditioning–speedup pair; eight are lost because Base errored. The whole-instance bootstrap uses 1185 complete variant–instance observations (seven of the possible  $298 \times 4$  lack the conditioning diagnostic). At 0.1%, 288 instances contribute; eighteen Base errors remove nine Simple, six SG-Ter-Mer, and three Full instances. Of 1152 potential rows, 1135 enter the joint comparison: twelve lack a usable variant speedup and five more lack the solver-conditioning diagnostic. Each of the 2000 resamples keeps all available formulation rows of an instance together. At 1%, the within-class solver-level coefficients are  $-0.31$  (Simple),  $-0.13$  (SG-Ter-Mer), and  $-0.12$  (Full); their varying magnitudes reinforce that the pooled coefficient is descriptive rather than an instance-level predictor.

*Exploratory dependence check.* If repeated rows and seeds are naively treated as independent observations, the first-order partial rank correlation between runtime improvement and the solver-facing conditioning diagnostic is approximately  $-0.25$  after controlling for loaded nonzeros, and a Williams–Steiger comparison gives  $Z \approx 6.1$ . Because both calculations ignore the clustered generator design, they are reported only as descriptive diagnostics and support no confirmatory claim.

**Table S7:** Available solver-internal-scaling sweep over both solvers, both MIPGap settings, and all three classes. Each cell is solver-time  $\text{sgm} (s = 1 \text{ s})/\text{PAR10}_{\text{avail}}$  in seconds, using the descriptive revision-campaign convention (timeouts receive 36000 s; pure pipeline errors are excluded). SA-Aug and SA-Mat retain the solver default. For Gurobi, Off/A/B/C mean  $\text{ScaleFlag}=0/1/2/3$ . For CPLEX they mean  $\text{Read}::\text{Scale}=-1/0/1/\text{not applicable}$ . The CPLEX Full 0.1%  $\text{Scale}=0$  artifact is partial (26 of 34 rows), and  $\text{Scale}=1$  is absent because the four-day sweep budget expired; all other configuration folders contain the manifest-declared number of rows. The lower panel audits the apparently fast CPLEX  $\text{Scale}=-1$  Full cell directly from the stored original-scale verifier fields.

| Panel A: timing sweep ( $\text{sgm}/\text{PAR10}_{\text{avail}}$ ) |      |            |                 |                 |                  |                 |                              |                 |                 |
|--------------------------------------------------------------------|------|------------|-----------------|-----------------|------------------|-----------------|------------------------------|-----------------|-----------------|
| Solver                                                             | Gap  | Class      | Base            | SA-Aug          | SA-Mat           | Off             | A                            | B               | C               |
| Gurobi                                                             | 1%   | Simple     | 3.936/11.26     | 2.133/2.50      | 2.166/2.56       | 41.969/1541.92  | 5.811/20.39                  | 4.402/12.60     | 3.791/10.79     |
| Gurobi                                                             | 1%   | SG-Ter-Mer | 14.087/564.33   | 8.282/554.10    | 8.635/554.64     | 56.625/740.17   | 14.664/566.35                | 20.907/589.78   | 21.475/580.46   |
| Gurobi                                                             | 1%   | Full       | 109.375/3448.17 | 95.239/3349.65  | 102.025/3458.10  | 561.945/9716.08 | 128.449/2444.67              | 138.112/2497.10 | 133.828/3468.97 |
| Gurobi                                                             | 0.1% | Simple     | 9.656/36.16     | 6.419/210.83    | 6.248/202.17     | 89.358/2031.10  | 15.299/219.91                | 10.607/33.11    | 9.438/35.10     |
| Gurobi                                                             | 0.1% | SG-Ter-Mer | 14.587/566.63   | 8.811/554.42    | 9.066/555.19     | 54.784/727.17   | 16.972/620.44                | 11.753/32.85    | 23.102/585.62   |
| Gurobi                                                             | 0.1% | Full       | 437.243/4766.52 | 388.446/4714.38 | 456.365/4821.25  | 396.978/9577.84 | 337.331/4621.00              | 347.613/3687.08 | 353.389/4697.74 |
| CPLEX                                                              | 1%   | Simple     | 2.621/7.24      | 1.725/3.65      | 1.861/4.01       | 2.500/3.93      | 2.549/7.59                   | 2.431/7.18      | –               |
| CPLEX                                                              | 1%   | SG-Ter-Mer | 2.253/3.57      | 2.400/10.26     | 2.309/5.94       | 5.168/12.29     | 2.229/3.53                   | 2.175/5.66      | –               |
| CPLEX                                                              | 1%   | Full       | 147.821/5609.27 | 127.360/3616.84 | 120.881/6509.40  | 14.381/35.45    | 159.012/5621.84              | 189.129/5644.99 | –               |
| CPLEX                                                              | 0.1% | Simple     | 5.143/215.88    | 4.266/188.91    | 4.450/203.68     | 7.634/209.70    | 5.203/216.59                 | 5.625/369.68    | –               |
| CPLEX                                                              | 0.1% | SG-Ter-Mer | 2.664/22.29     | 2.678/24.19     | 2.667/19.50      | 5.844/13.69     | 2.725/23.13                  | 2.477/11.78     | –               |
| CPLEX                                                              | 0.1% | Full       | 425.043/9419.41 | 426.712/9179.82 | 387.204/11235.11 | 13.476/24.97    | 303.376/6196.93 <sup>†</sup> | –               | –               |

| Panel B: CPLEX Full, $\text{Read}::\text{Scale} = -1$ , original-scale audit |      |             |            |                         |                       |                                                    |                                 |
|------------------------------------------------------------------------------|------|-------------|------------|-------------------------|-----------------------|----------------------------------------------------|---------------------------------|
| Gap                                                                          | Runs | Strict pass | Mixed pass | $\max v_{\text{row}}$   | $\max v_{\text{int}}$ | $\max  z_{\text{solver}} - z_{\text{recomputed}} $ | Strict-fail identifiers         |
| 1%                                                                           | 34   | 26          | 26         | $1.3149 \times 10^{-2}$ | 0                     | $3.73 \times 10^{-8}$                              | 002,003,004,012,014,022,023,044 |
| 0.1%                                                                         | 34   | 26          | 26         | $1.3149 \times 10^{-2}$ | 0                     | $3.73 \times 10^{-8}$                              | 002,003,004,012,014,022,023,044 |

*Notes.* The raw feasibility, integrality, and recomputed-objective fields are present for all 34 runs; the two gap folders contain identical objectives and verifier values. At 1%, 11 of 34  $\text{Scale}=-1$  objectives differ from the default-base objective by more than twice the stopping tolerance; six of those 11 also fail strict verification, while five pass it. Thus the speedup is real as timing but cannot be classified as a reliable solution improvement. The campaign’s generic `mip_equivalente` field labels every variant named `base` as `BASE`, irrespective of its internal-scaling parameter, so the pass counts above are rebuilt from the raw verifier fields rather than that label. <sup>†</sup>Partial artifact; its descriptive value is not comparable to a complete 34-instance cell.

**Table S8:** Archived runtime comparison for the operational coupling-stage activation campaign (Gurobi, 1% MIPGap). Solver-time shifted geometric mean (sgm,  $s = 1$  s) and descriptive  $\text{PAR10}_{\text{avail}}$  are seconds. The accompanying activation diagnostic found zero coupling-stage activations and byte-identical exports for each full/local-only pair; their small runtime differences therefore measure run-to-run variability, not a coupling-stage effect. The coupling-only variant coincides with Base at the exported-model level.

| Class      | Variant                 | sgm (s) | $\text{PAR10}_{\text{avail}}$ (s) |
|------------|-------------------------|---------|-----------------------------------|
| Simple     | Base                    | 3.73    | 10.47                             |
|            | SA-Aug (local+coupling) | 2.02    | 2.42                              |
|            | SA-Aug, local only      | 2.01    | 2.38                              |
|            | SA-Aug, coupling only   | 3.74    | 10.44                             |
|            | SA-Mat (local+coupling) | 2.04    | 2.43                              |
|            | SA-Mat, local only      | 2.06    | 2.48                              |
|            | SA-Mat, coupling only   | 3.73    | 10.26                             |
| Full       | Base                    | 82.55   | 3379.7                            |
|            | SA-Aug (local+coupling) | 71.41   | 3315.9                            |
|            | SA-Aug, local only      | 70.94   | 3314.4                            |
|            | SA-Aug, coupling only   | 83.69   | 3381.5                            |
|            | SA-Mat (local+coupling) | 78.75   | 4384.9                            |
|            | SA-Mat, local only      | 78.88   | 4386.1                            |
|            | SA-Mat, coupling only   | 90.21   | 3394.1                            |
| SG-Ter-Mer | Base                    | 14.42   | 565.6                             |
|            | SA-Aug (local+coupling) | 8.55    | 555.1                             |
|            | SA-Aug, local only      | 8.58    | 555.2                             |
|            | SA-Aug, coupling only   | 14.45   | 565.9                             |
|            | SA-Mat (local+coupling) | 8.96    | 555.6                             |
|            | SA-Mat, local only      | 8.83    | 555.6                             |
|            | SA-Mat, coupling only   | 14.43   | 565.8                             |

In SG-Ter-Mer one near-limit instance times out under every variant in this independent batch, so  $\text{PAR10}_{\text{avail}}$  is dominated by a single 36000-s penalty. In Full, three or four runs time out per variant, so the raw sgm values use different solved subsets and are descriptive. The attribution is the export/activation result: full equals local only, and coupling only equals Base.

**Table S9:** Direct SA–Flat reliability audit under Gurobi. Each row uses all available records from one dedicated class–tolerance batch; both SA and Flat were rerun in that batch. Timeouts (TO), strict/mixed original-scale verification failures (S/M), and fixed-pool PAR10 are reported for both coverage policies. Each PAR10 cell is ITT/strict/mixed in seconds, on the capped solver-time cost  $c^{(k, \text{solver})}$  of the main text; recomputing with the preprocessing-plus-solver cost shifts every cell by less than 0.2 s and changes no ordering.

| Gap  | Class      | Kernel      | $N$ | TO SA/Flat | SA fail S/M | Flat fail S/M | SA PAR10 I/S/M       | Flat PAR10 I/S/M     |
|------|------------|-------------|-----|------------|-------------|---------------|----------------------|----------------------|
| 1%   | Simple     | Augmented   | 204 | 0/0        | 12/12       | 13/12         | 2.4/2119.8/2119.8    | 2.4/2296.2/2119.7    |
|      | Simple     | Matrix-only | 204 | 0/0        | 19/18       | 19/18         | 2.4/3355.1/3178.7    | 2.4/3355.1/3178.6    |
|      | SG-Ter-Mer | Augmented   | 68  | 1/1        | 1/0         | 2/0           | 554.5/1083.9/554.5   | 541.1/1598.9/541.1   |
|      | SG-Ter-Mer | Matrix-only | 68  | 1/1        | 1/0         | 1/0           | 555.0/1083.7/555.0   | 551.6/1080.8/551.6   |
|      | Full       | Augmented   | 34  | 3/3        | 6/6         | 3/3           | 3312.1/9643.7/9643.7 | 3374.3/6464.9/6464.9 |
|      | Full       | Matrix-only | 34  | 3/3        | 2/2         | 4/4           | 3420.9/5533.6/5533.6 | 3430.6/7554.0/7554.0 |
| 0.1% | Simple     | Augmented   | 204 | 1/0        | 22/18       | 25/22         | 211.9/4090.0/3386.8  | 30.3/4436.5/3908.1   |
|      | Simple     | Matrix-only | 204 | 1/1        | 30/26       | 30/26         | 203.5/5491.9/4787.2  | 203.7/5492.1/4787.4  |
|      | SG-Ter-Mer | Augmented   | 68  | 1/1        | 1/0         | 1/0           | 556.0/1085.4/556.0   | 542.4/1071.4/542.4   |
|      | SG-Ter-Mer | Matrix-only | 68  | 1/1        | 1/0         | 1/0           | 556.5/1085.6/556.5   | 552.8/1082.0/552.8   |
|      | Full       | Augmented   | 30  | 3/3        | 2/2         | 2/1           | 3951.0/6342.2/6342.2 | 4004.3/6290.8/5109.3 |
|      | Full       | Matrix-only | 30  | 4/4        | 2/1         | 0/0           | 5193.8/7585.0/6389.5 | 5154.8/5154.8/5154.8 |

*Notes.* There are no algorithmic or pipeline aborts in these direct batches. Strict penalizes every strict original-scale failure; mixed penalizes only a failure that also lacks a reliable counterpart on the same instance, as defined in the main text. At Full 0.1%, instances 045–048 are absent from both policies. Under an equal-penalty imputation for those four unrecorded instances the ordering is unchanged and the added pairs are operational ties; that assumption does not address selection under unequal missing outcomes, for which the adversarial signed-rank sensitivity is reported in the main text and in the Notes to Tables S10–S12. The key reliability sensitivity is visible in Simple/Augmented: the small Flat solver-time advantage at 0.1% reverses under both strict and mixed costs. SG-Ter-Mer/Augmented retains a small descriptive Flat advantage because neither policy has a mixed failure.

**Table S10:** Magnitude and dependence sensitivities for the direct SA–Flat contrasts (exploratory endpoint hierarchy; see the main text). Panel A reports instance-level and operational-family inference for the two central 0.1% augmented contrasts on the prototype preprocessing-plus-solver implementation endpoint and on the capped solver-time decomposition, on the shared paired variable  $d_i = \log(T_{\text{Flat},i}/T_{\text{SA},i})$ , with the Hodges–Lehmann pseudomedian expressed as the typical SA/Flat ratio  $\exp(-\hat{\theta}_{\text{HL}})$ . Panel B reports the corresponding Gurobi Work-unit sensitivity. Panel C reports the own-incumbent path-specific diagnostic, which does not hold the integer solution fixed across policies.

| Panel A: paired log time ratio and dependence (central augmented 0.1% contrasts) |                     |        |              |            |                         |                       |                       |                       |                      |                      |                       |
|----------------------------------------------------------------------------------|---------------------|--------|--------------|------------|-------------------------|-----------------------|-----------------------|-----------------------|----------------------|----------------------|-----------------------|
| Class                                                                            | Endpoint            | $n/G$  | med. SA/Flat | HL SA/Flat | family-bootstrap 95% CI | $p_{\text{inst}}$     | $q_6$                 | $q_{12}$              | $p_{\text{block}}$   | $q_{6,\text{block}}$ | $q_{12,\text{block}}$ |
| Simple                                                                           | prototype pre+solve | 204/34 | 1.0320       | 1.0359     | [1.0274, 1.0418]        | $3.3 \times 10^{-11}$ | $2.0 \times 10^{-10}$ | $1.3 \times 10^{-10}$ | $5.1 \times 10^{-8}$ | $3.1 \times 10^{-7}$ | $2.1 \times 10^{-7}$  |
| Simple                                                                           | solver              | 204/34 | 1.0093       | 1.0135     | [1.0019, 1.0212]        | 0.00173               | 0.01038               | 0.02075               | 0.00266              | 0.01595              | 0.03190               |
| SG-Ter-Mer                                                                       | prototype pre+solve | 67/34  | 1.0473       | 1.1577     | [1.0170, 1.2065]        | 0.00031               | 0.00063               | 0.00063               | 0.00056              | 0.00111              | 0.00117               |
| SG-Ter-Mer                                                                       | solver              | 67/34  | 1.0159       | 1.1229     | [0.9797, 1.1309]        | 0.02616               | 0.05232               | 0.09346               | 0.03320              | 0.06640              | 0.11980               |

| Panel A2: prototype preprocessing-plus-solver endpoint, all twelve contrasts                                 |            |             |     |                       |                       |                       |                       |
|--------------------------------------------------------------------------------------------------------------|------------|-------------|-----|-----------------------|-----------------------|-----------------------|-----------------------|
| Adjusted values only; effect directions and median SA/Flat ratios are reported in Table 18 of the main text. |            |             |     |                       |                       |                       |                       |
| Gap                                                                                                          | Class      | Kernel      | $n$ | $q_6$                 | $q_{12}$              | $q_{6,\text{block}}$  | $q_{12,\text{block}}$ |
| 1%                                                                                                           | Simple     | Augmented   | 204 | $7.9 \times 10^{-26}$ | $1.6 \times 10^{-25}$ | $1.0 \times 10^{-9}$  | $2.1 \times 10^{-9}$  |
|                                                                                                              | Simple     | Matrix-only | 204 | $3.0 \times 10^{-32}$ | $6.0 \times 10^{-32}$ | $7.0 \times 10^{-10}$ | $1.4 \times 10^{-9}$  |
|                                                                                                              | SG-Ter-Mer | Augmented   | 67  | 0.00057               | 0.00063               | 0.00117               | 0.00117               |
|                                                                                                              | SG-Ter-Mer | Matrix-only | 67  | 0.01126               | 0.01287               | 0.01893               | 0.02163               |
|                                                                                                              | Full       | Augmented   | 31  | 0.1502                | 0.1502                | 0.1502                | 0.1502                |
|                                                                                                              | Full       | Matrix-only | 32  | 0.3403                | 0.3712                | 0.3403                | 0.3712                |
| 0.1%                                                                                                         | Simple     | Augmented   | 204 | $2.0 \times 10^{-10}$ | $1.3 \times 10^{-10}$ | $3.1 \times 10^{-7}$  | $2.1 \times 10^{-7}$  |
|                                                                                                              | Simple     | Matrix-only | 203 | $4.9 \times 10^{-8}$  | $4.9 \times 10^{-8}$  | 0.00042               | 0.00042               |
|                                                                                                              | SG-Ter-Mer | Augmented   | 67  | 0.00063               | 0.00063               | 0.00111               | 0.00117               |
|                                                                                                              | SG-Ter-Mer | Matrix-only | 67  | 0.02239               | 0.02468               | 0.02765               | 0.03072               |
|                                                                                                              | Full       | Augmented   | 27  | 0.02239               | 0.02487               | 0.02765               | 0.02798               |
|                                                                                                              | Full       | Matrix-only | 27  | 0.8408                | 0.8408                | 0.8408                | 0.8408                |

| Panel B: deterministic effort ( $\text{Work}_{\text{Flat}} - \text{Work}_{\text{SA}}$ ) |     |             |              |                |                 |                    |                      |                       |  |
|-----------------------------------------------------------------------------------------|-----|-------------|--------------|----------------|-----------------|--------------------|----------------------|-----------------------|--|
| Class                                                                                   | $n$ | Exact zeros | med. Work SA | med. Work Flat | $r_{\text{fb}}$ | $p_{\text{block}}$ | $q_{6,\text{block}}$ | $q_{12,\text{block}}$ |  |
| Simple                                                                                  | 204 | 138         | 2            | 2              | -0.382          | 0.01196            | 0.03587              | 0.04782               |  |
| SG-Ter-Mer                                                                              | 67  | 31          | 7            | 4              | -0.649          | 0.000970           | 0.00582              | 0.00637               |  |

**Panel C: median own-incumbent path-diagnostic ratio**

| Gap  | Class      | Kernel      | $n$ | med. $\rho_{\kappa,\text{SA}}^{\text{solver}}/\rho_{\kappa,\text{Flat}}^{\text{solver}}$ |
|------|------------|-------------|-----|------------------------------------------------------------------------------------------|
| 1%   | Simple     | Augmented   | 204 | 1.000                                                                                    |
|      | Simple     | Matrix-only | 204 | 1.000                                                                                    |
|      | SG-Ter-Mer | Augmented   | 67  | 0.626                                                                                    |
|      | SG-Ter-Mer | Matrix-only | 67  | 0.849                                                                                    |
|      | Full       | Augmented   | 31  | 0.628                                                                                    |
|      | Full       | Matrix-only | 32  | 0.849                                                                                    |
| 0.1% | Simple     | Augmented   | 204 | 1.000                                                                                    |
|      | Simple     | Matrix-only | 203 | 1.000                                                                                    |
|      | SG-Ter-Mer | Augmented   | 67  | 0.629                                                                                    |
|      | SG-Ter-Mer | Matrix-only | 67  | 0.849                                                                                    |
|      | Full       | Augmented   | 27  | 0.634                                                                                    |

**Table S11:** Attrition, original-batch preprocessing timing, and release-versus-diagnostic cached-metadata sensitivities for the direct SA–Flat controls. Panel D reports single-call preprocessing summaries from the original fixed-order solve batches. Panels E1–E2 use per-instance median preprocessing times estimated from the replicated low-contention protocols; uncertainty in those estimated medians is not propagated into the derived inferential quantities.

| Panel D: per-contrast attrition and preprocessing-time summaries |            |             |         |          |             |                  |          |                                       |                                         |               |
|------------------------------------------------------------------|------------|-------------|---------|----------|-------------|------------------|----------|---------------------------------------|-----------------------------------------|---------------|
| Gap                                                              | Class      | Kernel      | Nominal | Recorded | Common pool | Double noncompl. | $n$ used | med. $T_{\text{pre}}^{\text{SA}}$ (s) | med. $T_{\text{pre}}^{\text{Flat}}$ (s) | med. diff (s) |
| 1%                                                               | Simple     | Augmented   | 204     | 204      | 204         | 0                | 204      | 0.074                                 | 0.025                                   | +0.050        |
|                                                                  | Simple     | Matrix-only | 204     | 204      | 204         | 0                | 204      | 0.077                                 | 0.025                                   | +0.053        |
|                                                                  | SG-Ter-Mer | Augmented   | 68      | 68       | 68          | 1                | 67       | 0.049                                 | 0.019                                   | +0.031        |
|                                                                  | SG-Ter-Mer | Matrix-only | 68      | 68       | 68          | 1                | 67       | 0.051                                 | 0.019                                   | +0.033        |
|                                                                  | Full       | Augmented   | 34      | 34       | 34          | 3                | 31       | 0.121                                 | 0.041                                   | +0.080        |
|                                                                  | Full       | Matrix-only | 34      | 34       | 34          | 2                | 32       | 0.125                                 | 0.041                                   | +0.084        |
| 0.1%                                                             | Simple     | Augmented   | 204     | 204      | 204         | 0                | 204      | 0.077                                 | 0.025                                   | +0.051        |
|                                                                  | Simple     | Matrix-only | 204     | 204      | 204         | 1                | 203      | 0.079                                 | 0.025                                   | +0.054        |
|                                                                  | SG-Ter-Mer | Augmented   | 68      | 68       | 68          | 1                | 67       | 0.051                                 | 0.019                                   | +0.032        |
|                                                                  | SG-Ter-Mer | Matrix-only | 68      | 68       | 68          | 1                | 67       | 0.052                                 | 0.019                                   | +0.033        |
|                                                                  | Full       | Augmented   | 34      | 30       | 30          | 3                | 27       | 0.120                                 | 0.041                                   | +0.079        |
|                                                                  | Full       | Matrix-only | 34      | 30       | 30          | 3                | 27       | 0.124                                 | 0.041                                   | +0.083        |

**Panel E1: release-build cached-metadata endpoint (implementation-cost sensitivity of record)**

| Gap  | Class             | Kernel | $n$ | med. SA/Flat | $r_{\text{rb}}$ | $q_6$ | $q_{12}$ | $q_{6,\text{block}}$ | $q_{12,\text{block}}$ |
|------|-------------------|--------|-----|--------------|-----------------|-------|----------|----------------------|-----------------------|
| 1%   | Simple            | Aug    | 204 | 1.0004       | −0.090          | 0.318 | 0.364    | 0.337                | 0.450                 |
|      | Simple            | Mat    | 204 | 1.0011       | −0.167          | 0.115 | 0.092    | 0.173                | 0.162                 |
|      | SG-Ter-Mer        | Aug    | 67  | 1.0075       | −0.303          | 0.115 | 0.092    | 0.173                | 0.120                 |
|      | SG-Ter-Mer        | Mat    | 67  | 0.9976       | −0.140          | 0.318 | 0.381    | 0.337                | 0.450                 |
|      | Full <sup>†</sup> | Aug    | 31  | 0.9193       | +0.355          | 0.173 | 0.173    | 0.173                | 0.173                 |
|      | Full <sup>†</sup> | Mat    | 32  | 0.9866       | +0.246          | 0.318 | 0.364    | 0.337                | 0.396                 |
| 0.1% | Simple            | Aug    | 204 | 1.0094       | −0.253          | 0.010 | 0.021    | 0.016                | 0.031                 |
|      | Simple            | Mat    | 203 | 0.9981       | +0.089          | 0.410 | 0.364    | 0.546                | 0.469                 |
|      | SG-Ter-Mer        | Aug    | 67  | 1.0159       | −0.313          | 0.051 | 0.092    | 0.065                | 0.120                 |
|      | SG-Ter-Mer        | Mat    | 67  | 1.0008       | −0.109          | 0.526 | 0.478    | 0.546                | 0.496                 |
|      | Full <sup>†</sup> | Aug    | 27  | 0.8771       | +0.529          | 0.045 | 0.090    | 0.045                | 0.090                 |
|      | Full <sup>†</sup> | Mat    | 27  | 0.9803       | +0.053          | 0.822 | 0.822    | 0.822                | 0.822                 |

<sup>†</sup> Approximate cross-build sensitivity; five Full instances exhibit last-bit model-construction differences. See Section S2.1.

**Panel E2: non-optimized diagnostic-build audit**

| Gap  | Class      | Kernel | $n$ | med. SA/Flat | $r_{\text{rb}}$ | $q_6$  | $q_{12}$ | $q_{6,\text{block}}$ | $q_{12,\text{block}}$ |
|------|------------|--------|-----|--------------|-----------------|--------|----------|----------------------|-----------------------|
| 1%   | Simple     | Aug    | 204 | 1.0008       | −0.124          | 0.185  | 0.212    | 0.258                | 0.295                 |
|      | Simple     | Mat    | 204 | 1.0019       | −0.333          | 0.0002 | 0.0005   | 0.004                | 0.008                 |
|      | SG-Ter-Mer | Aug    | 67  | 1.0074       | −0.303          | 0.093  | 0.075    | 0.119                | 0.095                 |
|      | SG-Ter-Mer | Mat    | 67  | 0.9976       | −0.148          | 0.291  | 0.388    | 0.310                | 0.414                 |
|      | Full       | Aug    | 31  | 0.9197       | +0.355          | 0.173  | 0.173    | 0.173                | 0.173                 |
|      | Full       | Mat    | 32  | 0.9869       | +0.242          | 0.286  | 0.358    | 0.286                | 0.358                 |
| 0.1% | Simple     | Aug    | 204 | 1.0096       | −0.259          | 0.008  | 0.008    | 0.013                | 0.013                 |
|      | Simple     | Mat    | 203 | 0.9983       | +0.065          | 0.506  | 0.460    | 0.633                | 0.576                 |
|      | SG-Ter-Mer | Aug    | 67  | 1.0159       | −0.314          | 0.051  | 0.075    | 0.065                | 0.095                 |
|      | SG-Ter-Mer | Mat    | 67  | 1.0036       | −0.116          | 0.506  | 0.460    | 0.633                | 0.512                 |
|      | Full       | Aug    | 27  | 0.8771       | +0.529          | 0.045  | 0.060    | 0.045                | 0.060                 |

## Section S2.1. Timing, build, and transfer-factor audits

**Table S12: timing, build, and cross-environment transfer-factor audits for the direct SA–Flat controls.** Panel E3 reports the number and identity of fast-class contrasts with  $q_6 < 0.05$  over the hypothetical transfer-factor grids; Panel E4 reports the corresponding Full-Augmented sensitivity; Panel E5 reports the release-build absolute preprocessing times. The accompanying text documents the build comparison and the export-invariance audit. The transfer factors are hypothetical: the workstation-to-ClusterUY transfer was never observed, and the only measured perturbation is the change of toolchain on the same machine.

† Five Full instances exhibit last-bit model-construction differences, so the Full release-cached values in Panel E1 should be read as approximate build sensitivities rather than exact recombinations of one model; see the build-audit details below and `results-revision/preproc-timing/build_invariance.csv`. On the cached counterpart, the role-blind rule has the lower per-instance median in 585/612 records under the pinned release build, 607/612 under the pinned diagnostic build and 408/612 under the concurrent diagnostic protocol. For a descriptive within-instance dispersion comparison, we call record  $i$  a *within-dispersion tie* for build  $b$  when

$$|\tilde{T}_{\text{preproc,SA},i}^{\text{cached},b} - \tilde{T}_{\text{preproc,Flat},i}^{\text{cached},b}| \leq B_i^b, \quad B_i^b = \max\left\{\text{IQR}(T_{\text{preproc,SA},i}^{\text{cached},b}), \text{IQR}(T_{\text{preproc,Flat},i}^{\text{cached},b})\right\},$$

where  $\tilde{T}$  denotes the median and the interquartile ranges are taken over the  $R = 20$  repetitions; the quantities are cached *preprocessing* times, not the preprocessing-plus-solver endpoint. Using this descriptive scale-of-dispersion classification, the three-way classification (role-blind-favoured, selective-favoured, tie) is 461/1/150 under the pinned release build, 554/0/58 under the pinned diagnostic build and 37/0/575 under the concurrent protocol; each row sums to 612. The band is an ad hoc scale-of-dispersion heuristic based on the larger marginal IQR; it is not an uncertainty estimate for the paired median difference, nor a confidence interval, an equivalence test or a domain-calibrated practical-significance margin, so records inside it are within-dispersion ties rather than statistically indeterminate. A sharper screen would use the per-repetition paired differences  $D_{ir}^b$  and their  $\text{IQR}_r$ , which the archived summaries do not retain. These counts condition on the  $R = 20$  per-instance median estimates and do not propagate within-instance timing uncertainty. The three record-level counts are computed by `scripts/preproc_timing_replication.py` from the `cached_sa_s` and `cached_flat_s` columns of `results-revision/preproc-timing/replicated_pinned_release.csv` (pinned release), `results-revision/preproc-timing/replicated_pinned_diagnostic.csv` (pinned diagnostic) and `results-revision/preproc-timing/replicated.csv` (concurrent diagnostic).

**Panel E3: release-build cross-environment transfer-factor sensitivity**

| Grid point                                               | Fast-class contrasts with $q_6 < 0.05$ (of eight) |     |   |   |   |   |
|----------------------------------------------------------|---------------------------------------------------|-----|---|---|---|---|
| Common $\lambda$                                         | 0.25                                              | 0.5 | 1 | 2 | 4 | 8 |
|                                                          | 1                                                 | 1   | 1 | 1 | 2 | 2 |
| Ratio $\rho = \lambda_{\text{SA}}/\lambda_{\text{Flat}}$ | 0.25                                              | 0.5 | 1 | 2 | 4 | — |
|                                                          | 1                                                 | 1   | 1 | 3 | 5 | — |

### Measurement and build-audit details.

The crossing contrasts are identified as follows. On the common-factor grid, Simple/Augmented at 0.1% crosses at every point; Simple/Matrix-only at 1% joins it at  $\lambda = 4$  and 8. On the policy-specific grid, Simple/Augmented at 0.1% crosses at every point; at  $\rho = 2$  it is joined by SG-Ter-Mer/Augmented at 0.1% and Simple/Matrix-only at 1%; at  $\rho = 4$  also by SG-Ter-Mer/Augmented at 1% and Simple/Augmented at 1%. Panel E3 is restricted to the eight fast-class contrasts. In the separate Full-Augmented 0.1% calculation,  $q_6 = 0.045$  at every evaluated common-factor point and at the policy-specific points  $\rho \leq 2$ , falling to 0.036 at  $\rho = 4$ ; the joint value  $q_{12}$  remains above 0.05 except at  $\rho = 4$ , where it is 0.042 (Panel E4). None of these nominal Full crossings is retained by the adversarial missing-pair sensitivity. The  $q_{12} = 0.042$  crossing at the extreme evaluated point  $\rho = 4$  is moreover conditional on a hypothetical policy-specific transfer factor, on approximate cross-build Full models, and on the 27 observed eligible pairs; it is not treated as evidence of a robust Full coverage ordering. No other Full contrast has  $q_6 < 0.05$  anywhere on the grids.

**Panel E4: Full-Augmented at 0.1% across the transfer grids** (approximate cross-build row;  $n = 27$ ).

|                  |       |       |       |       |              |       |
|------------------|-------|-------|-------|-------|--------------|-------|
| Common $\lambda$ | 0.25  | 0.5   | 1     | 2     | 4            | 8     |
| $q_6$            | 0.045 | 0.045 | 0.045 | 0.045 | 0.045        | 0.045 |
| $q_{12}$         | 0.090 | 0.090 | 0.090 | 0.063 | 0.060        | 0.060 |
| Ratio $\rho$     | 0.25  | 0.5   | 1     | 2     | 4            | —     |
| $q_6$            | 0.045 | 0.045 | 0.045 | 0.045 | 0.036        | —     |
| $q_{12}$         | 0.090 | 0.090 | 0.090 | 0.060 | <b>0.042</b> | —     |

The adversarial missing-pair sensitivity was recomputed at every grid point on this cached endpoint: its effective nonzero sample remains  $n = 31$  and  $p_{\text{adv}} = 0.433$  at all eleven points, so no point falls below 0.05 (the corresponding value on the prototype endpoint is 0.47). For the observed-pair, non-adversarial analysis, the unadjusted values are likewise invariant along both grids ( $p = 0.0151$ ,  $p_{\text{block}} = 0.0151$ ), because scaling the preprocessing term cannot change the paired ranks here—that term is milliseconds against solver times of seconds—so only the BH-adjusted values move, and they move because the other contrasts in the family do. Because every Full instance is a singleton operational family, the family-level values coincide with the instance-level values at every grid point:  $q_{6,\text{block}} = q_6$  and  $q_{12,\text{block}} = q_{12}$ .

The release-build endpoint (Panel E1) is the implementation-cost sensitivity used in the main text; the diagnostic-build results (Panel E2) are retained only to show the sensitivity of the cached comparison to compiler optimization. Panel E3 reports within-tolerance  $q_6$  crossings only; these counts do not imply retention under the joint or operational-family adjustments. Among the eight fast-class contrasts summarized in Panel E3, Simple-Augmented at 0.1% is the only contrast with  $q_6 < 0.05$  at every evaluated grid point; the separate approximate Full-Augmented row also has  $q_6 < 0.05$  throughout, but its evidence is conditional on the observed eligible pairs and is not retained by the adversarial missing-pair sensitivity. Additional within-tolerance  $q_6$  crossings first appear at the evaluated point  $\rho = 2$ ; the exact transition threshold between 1 and

2 is not identified by this grid. The transfer factors are hypothetical: the workstation-to-ClusterUY transfer was never observed, and the only measured perturbation is the change of toolchain on the same machine. Comparing Panels E1 and E2 gives the implied toolchain ratio. Writing, for build  $b \in \{\text{release, diagnostic}\}$ , class  $c$  and kernel  $k$ ,

$$R_{b,c,k} = \exp\left[\text{median}_i \log\left(T_{\text{preproc,SA},i}^{\text{cached},b} / T_{\text{preproc,Flat},i}^{\text{cached},b}\right)\right], \quad \rho_{c,k}^{\text{toolchain}} = \frac{R_{\text{release},c,k}}{R_{\text{diagnostic},c,k}},$$

The quantities in this definition are cached *preprocessing* times only, not the cached preprocessing-plus-solver endpoint of the main-text table; it is a ratio of exponentiated median log-ratios, not a ratio of marginal median preprocessing times (the quantity computed by `scripts/cached_transfer_sensitivity.py`). Its values are 0.947, 0.942, 1.012, 0.994, 0.978 and 0.966 for Full/Aug, Full/Mat, SG-Ter-Mer/Aug, SG-Ter-Mer/Mat, Simple/Aug and Simple/Mat; optimization leaves the prototype preprocessing-call ratios essentially unchanged (5.38–6.12× against 5.07–6.00×) and the metadata share at 81–83%. Exporting the preprocessed model with both builds over all 306 instances, both kernels and both coverage policies gives 1204/1224 byte-identical exports (`scripts/build_invariance_audit.py`; full record in `results-revision/preproc-timing/build_invariance.csv`). All Simple and SG-Ter-Mer records are identical. The 20 exceptions are five Full instances under all four variants, where the nonlinear-hydro linearization coefficients differ by one unit in the last place ( $4.7 \times 10^{-16}$  relative); the audit localizes the discrepancy to optimization-dependent floating-point evaluation during model construction rather than to a change in the selected scaling factors or safeguard decisions. The complete contrast-by-grid values—every contrast, grid point, median ratio,  $r_{\text{rb}}$ ,  $p$ ,  $q_6$  and  $q_{12}$ —are in `results-revision/preproc-timing/cached_transfer_sensitivity.csv` of the reproducibility package, regenerated by `scripts/cached_transfer_sensitivity.py --dump-csv`.

**Panel E5: release-build absolute preprocessing times**, in milliseconds as median [IQR]; an IQR below 0.001 ms is shown as < 0.001 rather than rounded to zero. The absolute times quoted elsewhere in these notes are from the diagnostic build.

| Class      | Kernel      | Prototype preprocessing call (ms) |               | Cached preprocessing counterpart (ms) |               |
|------------|-------------|-----------------------------------|---------------|---------------------------------------|---------------|
|            |             | SA                                | Flat          | SA                                    | Flat          |
| Simple     | Augmented   | 8.447 [0.093]                     | 1.410 [0.044] | 1.362 [0.012]                         | 1.288 [0.043] |
| Simple     | Matrix-only | 8.606 [0.100]                     | 1.405 [0.048] | 1.528 [0.011]                         | 1.283 [0.044] |
| SG-Ter-Mer | Augmented   | 5.335 [0.041]                     | 0.991 [0.005] | 0.911 [0.006]                         | 0.894 [0.004] |
| SG-Ter-Mer | Matrix-only | 5.433 [0.040]                     | 0.975 [0.006] | 1.006 [0.009]                         | 0.877 [0.006] |
| Full       | Augmented   | 13.607 [0.121]                    | 2.401 [0.119] | 2.248 [0.044]                         | 2.220 [0.121] |
| Full       | Matrix-only | 13.878 [0.147]                    | 2.384 [0.085] | 2.514 [0.093]                         | 2.203 [0.077] |

*Notes to Tables S10–S12.* Both time endpoints use the paired variable  $d_i = \log(T_{\text{Flat},i}/T_{\text{SA},i})$ ; negative values favor Flat. The test, the effect size, and the median ratio are computed on this one scale and share its direction convention but do not target an identical estimand; the Hodges–Lehmann column is the location estimate aligned with the signed-rank test.  $G$  is the number of operational families. Simple groups the six `casoXX[a–f]` scenarios with a common root; SG-Ter-Mer groups each `instanceXXX/instance(XXX+48)` pair (34 pairs, no singletons); Full is singleton by construction. The exact block-sign test preserves each scenario’s absolute-rank contribution, sums those contributions within family, and flips one common sign per family; its null distribution is computed by dynamic programming. The percentile bootstrap resamples whole families ( $B = 2000$ , seed 12345; the pre-solve and solver-time intervals are drawn from one RNG stream in that order). Negative Work  $r_{\text{rb}}$  favors Flat under the sign convention of the main table. Work reproduces both augmented 0.1% solver-time directions but, with medians 2 versus 2 in Simple and 7 versus 4 in SG-Ter-Mer, does not turn the small solver-time effects into large speedups; a single default-seed run per cell means seed- and trajectory-level variability is not eliminated. In Panel C each policy fixes the integer variables to the incumbent found by that same run. The ratio can therefore reflect both representation and a different integer solution; no common-incumbent diagnostic can be reconstructed from the archived artifacts. In Panel D, “Recorded” counts instances present in the batch CSV (at Full 0.1%, instances 045–048 were not recorded), the common pool removes uniformly unavailable instances (none here beyond the unrecorded ones), and double noncompletions are pairs in which neither policy completes. No pair is dropped for a missing time and no exact-zero difference occurs;  $n$  used applies to both endpoints. In these batches the preprocessing calls are timed once each, with the C++ high-resolution wall clock, in the batch’s fixed (un-randomized) execution order and without repetitions; per-policy IQRs are at most 0.003 s in every cell. Timing environment for the preprocessing replication: Intel Core i7-13620H (10 physical / 16 logical cores, 0.4–4.9 GHz, powersave governor), Ubuntu 22.04.5 with Linux 6.8.0, g++ 11.4.0 at `-std=c++17 -g (no -O flag)`, binary code/build\_diag/SistemaElectrico; the pinned protocol binds one worker to a single logical CPU with `taskset`, the concurrent protocol uses 12 unpinned workers. The release-build replication uses the same source commit, hardware, operating system, compiler, libraries, timing protocol, seeds, execution order and affinity policy, compiled with `-std=c++17 -O2 -DNDEBUG` into code/build\_release/SistemaElectrico; the two builds differ only in optimization flags. The solve batches ran on ClusterUY as described in the main text, so the cached-metadata endpoint is a cross-protocol counterfactual.

*Order-randomized preprocessing replication* (`scripts/preproc_timing_replication.py`;  $R = 20$ , seeded order per repetition, fresh process per call, MIP never solved). Two protocols were run: a *concurrent* stress test (12 workers on a 16-core workstation, no CPU pinning) and a *low-contention* estimate (one worker, every call pinned to one logical CPU via `--pin-cpu`). The pinned protocol is the cleaner estimator—relative IQR 0.00–0.02 against 0.18–1.83—and gives, per class and kernel, selective versus role-blind medians and paired ratios: Simple/Aug 0.0487 vs 0.0083 (5.85×; cached 0.0087 vs 0.0081, 1.08×); Simple/Mat 0.0499 vs 0.0083 (6.00×; cached 0.0099 vs 0.0081, 1.23×); SG-Ter-Mer/Aug 0.0300 vs 0.0059 (5.07×; cached 0.0058 vs 0.0057, 1.01×); SG-Ter-Mer/Mat 0.0307 vs 0.0058 (5.28×; cached 0.0065 vs 0.0056, 1.15×); Full/Aug 0.0773 vs 0.0136 (5.71×; cached 0.0141 vs 0.0132, 1.07×); Full/Mat 0.0797 vs 0.0137 (5.81×; cached 0.0161 vs 0.0133, 1.21×). All 612 instance–kernel records favor the role-blind rule on the full call under both protocols. On the cached counterpart the role-blind rule has the lower median in 20–32 of the 34 operational families per cell, so that residual is descriptive rather than uniform.

The concurrent protocol is retained as a stress test of the direction under adverse measurement conditions. Its per-class and per-kernel values—medians of the per-instance median call time (selective versus role-blind, seconds), the median paired ratio, the cached-metadata counterpart  $T_{\text{diagnosis}} + T_{\text{application}}$ , and the relative interquartile range, which replaces the coefficient of variation because the latter is unstable for short, skewed timings—are: Simple/Aug 0.272 vs 0.031 (8.38×; cached 0.032 vs 0.029, 1.08×; relative IQR 0.32 vs 1.51); Simple/Mat 0.281 vs 0.031 (8.58×; cached 0.040 vs 0.029, 1.40×; 0.32 vs 1.58); SG-Ter-Mer/Aug 0.165 vs 0.018 (9.10×; cached 0.020 vs 0.018, 1.08×; 0.20 vs 1.70); SG-Ter-Mer/Mat 0.167 vs 0.020 (8.54×; cached 0.023 vs 0.019, 1.24×; 0.18 vs 1.83); Full/Aug 0.412 vs 0.066 (6.45×; cached 0.072 vs 0.062, 1.15×; 0.21 vs 1.15); Full/Mat 0.424 vs 0.072 (6.08×; cached 0.084 vs 0.067, 1.32×; 0.23 vs 1.04). The selective median exceeds the role-blind median on all 612 instance–kernel records (paired Wilcoxon  $p \leq 1.2 \times 10^{-10}$  per cell), and every operational family—34 in each class—has a positive median difference in both kernels. Metadata reconstruction accounts for 72–89% of the selective preprocessing time and at most 0.6 ms of the role-blind time. The batches contain no one-sided aborts, so the  $10T_{\text{max}}$  abort sensitivity is identical to the primary convention. Each of the six dedicated CSVs contains both SA and Flat reruns from the same class–tolerance batch, independent of the main campaign. Adversarial signed-rank sensitivity for the four unrecorded pairs (Full augmented 0.1%, pre+solve endpoint): appending instances 045–048 with the adverse sign at above-maximal ranks gives  $p = 0.47$  and  $r_{\text{tb}} = +0.15$ . For this sensitivity the nominal pool contains all 34 instances—the 27 observed nonzero pairs, the four appended adverse missing pairs and the three double timeouts restored as exact-zero operational ties—and the signed-rank procedure discards those three zeros, leaving an effective nonzero sample of  $n = 31$ , so the adjusted Full contrast is conditional on the 27 eligible observed pairs. This construction bounds the signed-rank evidence under adverse missing pairs; it is not a universal worst-case bound for the median ratio, the Hodges–Lehmann estimate, or the reliability-adjusted aggregate costs. On the same augmented sample the adversarial location estimates remain directionally favorable to selectivity (median SA/Flat 0.9596, HL 0.9465); the matrix-only counterpart moves the other way (median 1.0861, HL 1.0977,  $p = 0.34$ ). The three double timeouts enter as operational ties (exact zeros).

## Section S3. Traceability of the kernel-matched metadata grid

This section documents the experiment behind the kernel-matched metadata grid of the main text (the comparison of the matrix-only structure-aware rule SA-Mat against a same-kernel role-blind control), so the paper’s central positive finding is fully reproducible.

**Instance family.** Every cell uses the block-heterogeneous coefficient family on  $B = 4$  blocks, each with 6 continuous and 2 binary variables, 8 local rows and 3 coupling rows. Block  $i$  (1-indexed) carries a base magnitude  $10^{e_i}$  with the exponents spread uniformly over  $[-R, R]$ ,  $e_i = -R + 2R(i - 1)/(B - 1)$ , so that equal absolute coefficients denote different physical scales across blocks. The block-scale range  $R$  is the row variable of the grid and is set through the SYNTH\_BLOCK\_RANGE environment variable (default 2, which reproduces the original block-heterogeneous set byte-for-byte). At  $R = 2$  the four block magnitudes are  $10^{-2}$ ,  $10^{-2/3}$ ,  $10^{2/3}$ ,  $10^2$ ; at  $R = 3$  they are  $10^{-3}$ ,  $10^{-1}$ ,  $10^1$ ,  $10^3$ . The coupling disturbance of severity  $S$  is applied on top of this, as in the main synthetic grid.

**Seeds and variants.** Each cell draws 30 instances from consecutive seeds starting at 20250803. The variants are Base, the matrix-only structure-aware rule SA-Mat, and its kernel-matched role-blind control Flat-Mat (the flat rule run with the coefficient-only kernel, added to the synthetic runner for this test); SA-Aug, Flat and Ruiz are included on the headline cell for reference. The exported constraint matrix of every variant is read back and decomposed with `scripts/spectral_audit.py`, and the reported statistic is the paired ratio  $\kappa_2^+(SA - Mat)/\kappa_2^+(\text{Flat-Mat})$ ; because the two rules share both kernel and coverage, the ratio isolates the block metadata.

**Headline cell.** Table S13 gives the per-variant  $\kappa_2^+$  distribution on the headline cell (coupling pattern,  $R = 2$ ,  $S = 6$ , 30 seeds). The matrix-only structure-aware rule is lower than its role-blind twin on all 30 instances; the homogeneous control is the  $[-1, 1]$  row of the metadata grid in the main text, where the two coincide within noise.

**Table S12:** Per-variant  $\kappa_2^+$  on the headline cell of the kernel-matched grid (coupling pattern, block-scale range  $[-2, 2]$ ,  $S = 6, 30$  seeds): median, interquartile range, and full range. The paired SA-Mat/Flat-Mat ratio has median 0.098 (30/30 instances, Wilcoxon  $p = 1.9 \times 10^{-9}$ ).

| Variant  | Median $\kappa_2^+$ | IQR                                  | Range                                   |
|----------|---------------------|--------------------------------------|-----------------------------------------|
| Base     | $5.3 \times 10^8$   | $[2.4 \times 10^7, 3.2 \times 10^9]$ | $[4.2 \times 10^5, 2.2 \times 10^{11}]$ |
| SA-Mat   | $1.3 \times 10^3$   | $[1.2 \times 10^3, 1.7 \times 10^3]$ | $[8.7 \times 10^2, 1.7 \times 10^4]$    |
| Flat-Mat | $1.4 \times 10^4$   | $[1.1 \times 10^4, 1.9 \times 10^4]$ | $[6.2 \times 10^3, 2.1 \times 10^5]$    |
| SA-Aug   | $2.1 \times 10^5$   | $[4.3 \times 10^4, 6.1 \times 10^5]$ | $[1.2 \times 10^4, 1.1 \times 10^7]$    |
| Flat     | $2.4 \times 10^4$   | $[1.8 \times 10^4, 3.1 \times 10^4]$ | $[8.5 \times 10^3, 3.4 \times 10^5]$    |
| Ruiz     | $3.9 \times 10^5$   | $[8.3 \times 10^4, 9.2 \times 10^5]$ | $[2.6 \times 10^4, 1.3 \times 10^7]$    |

**Reproduction.** The full 18-cell sweep and the per-instance  $\kappa_2^+$  values are archived under `results-revision/spectral` (`metadata_grid.csv` for the sweep, `coupling_blockhet_spectral.csv` for the headline cell). One cell is produced by setting `SYNTH_BLOCK_RANGE=2` and piping to the solver the command `benchmarkSintetico --coef-family block_heterogeneous --volcar-lp <dir>` followed by the parameter lines `DummyLp, seed 20250803, 30 instances, 4 blocks, 6 2, 8 3, pattern coupling, severity 6, variant list Base, SA-Aug, SA-Mat, Flat,` and an output path; then `scripts/spectral_audit.py --lp-dir <dir>` reads the exported matrices. The whole grid is driven by `scripts/synth_metadata_grid.py`.

## Section S4. Algorithmic summary

The recommended rule is the exhaustive matrix-only structure-aware scaling of Algorithm S1; the augmented variant studied for comparison differs only in folding the right-hand side into the magnitude extremes, as noted after it. Equation numbers refer to the main text.

Both assume an *exhaustive* role map—every row is either local to one block or a coupling row,  $\mathcal{R}_{\text{all}} = (\dot{\bigcup}_{i \in \mathcal{B}} \mathcal{R}_i) \dot{\cup} \mathcal{R}_{\text{cpl}}$ , with no residual category. The archived operational campaigns used a non-exhaustive implementation in which a block of market-shortage rows remained unassigned and therefore unscaled; Algorithm S1 instead states the exhaustive rule that defines the target method, and the coverage-restored control of the main text removes the confound that non-exhaustive map introduced.

**Augmented exploratory variant (SA-Aug).** The augmented rule of the comparison is obtained from Algorithm S1 by a single change: the per-row set  $S_r^A$  and the block extremes  $M_i, m_i$  are formed from the nonzero coefficients *and* the nonzero right-hand side, i.e. their  $A, b$  counterparts (Eqs. (16), (20), (22) of the main text), with an optional local reliability floor  $\varepsilon_{\text{floor}}^{\text{loc}}$  gating the local update. Folding  $|b_r|$  into the extremes is what distinguishes SA-Aug; it is the variant that, under the strict absolute criterion, carries the additional documented original-scale reliability failure mode on zero-right-hand-side balance rows, and it obtains no spectral advantage over its role-blind control. We therefore report it as exploratory and recommend the matrix-only rule above.

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**Algorithm S1** Exhaustive matrix-only structure-aware row scaling (recommended rule)

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**Require:** Matrix  $A$ , right-hand side  $b$ , exhaustive row roles (local/coupling), block ownership  $\beta(\cdot)$ , coupling incidence

**Ensure:** Scaled matrix  $\tilde{A}$ , scaled right-hand side  $\tilde{b}$

- 1: Initialize row factors  $d_r \leftarrow 1$  for every row
  - 2: **for** each local block  $i \in \mathcal{B}$  **do**
  - 3:     **for** each local row  $r \in \mathcal{R}_i$  **do**
  - 4:         Build  $S_r^A$  from the nonzero coefficients of row  $r$  (the right-hand side is excluded)
  - 5:         **if**  $S_r^A \neq \emptyset$  **then**
  - 6:             Compute  $M_r^A = \max S_r^A$  and  $m_r^A = \min S_r^A$
  - 7:             Set  $\alpha_r^A \leftarrow 1/\sqrt{M_r^A m_r^A}$
  - 8:             **if** the safeguards admit  $\alpha_r^A$  **then**
  - 9:                 Update  $d_r \leftarrow d_r \alpha_r^A$
  - 10:             **end if**
  - 11:         **end if**
  - 12:     **end for**
  - 13: **end for**
  - 14: Compute the locally scaled coefficients  $\tilde{a}_{r,j} = d_r a_{r,j}$
  - 15: Compute block scales  $s_i^A = \sqrt{M_i^A m_i^A}$  from the block coefficient extremes
  - 16: Compute  $\hat{\rho}_r^A$  for every coupling row (its coefficients against the incident block scales  $s_i^A$ ); set  $\rho \leftarrow \max_r \hat{\rho}_r^A$
  - 17: **if**  $\rho$  exceeds the coupling threshold **then**
  - 18:     Set  $\gamma^A \leftarrow \arg \min_{\gamma} \Phi(\gamma)$  over the logarithmic grid  $\gamma \in [10^{-6}, 10^6]$ ; clip  $\gamma^A$  to  $[10^{-8}, 10^8]$
  - 19:     **if**  $\gamma^A$  lies outside the near-identity band **then**
  - 20:         **for** each coupling row  $r$  **do**
  - 21:             **if**  $\gamma^A$  keeps row  $r$  inside the coefficient window **and** above the reliability floor **then**
  - 22:                 Update  $d_r \leftarrow d_r \gamma^A$
  - 23:             **end if**
  - 24:         **end for**
  - 25:     **end if**
  - 26: **end if**
  - 27: Return  $\tilde{A} = \text{diag}(d)A$  and  $\tilde{b} = \text{diag}(d)b$
-